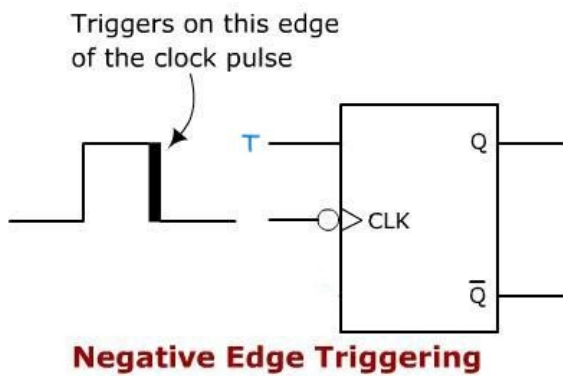
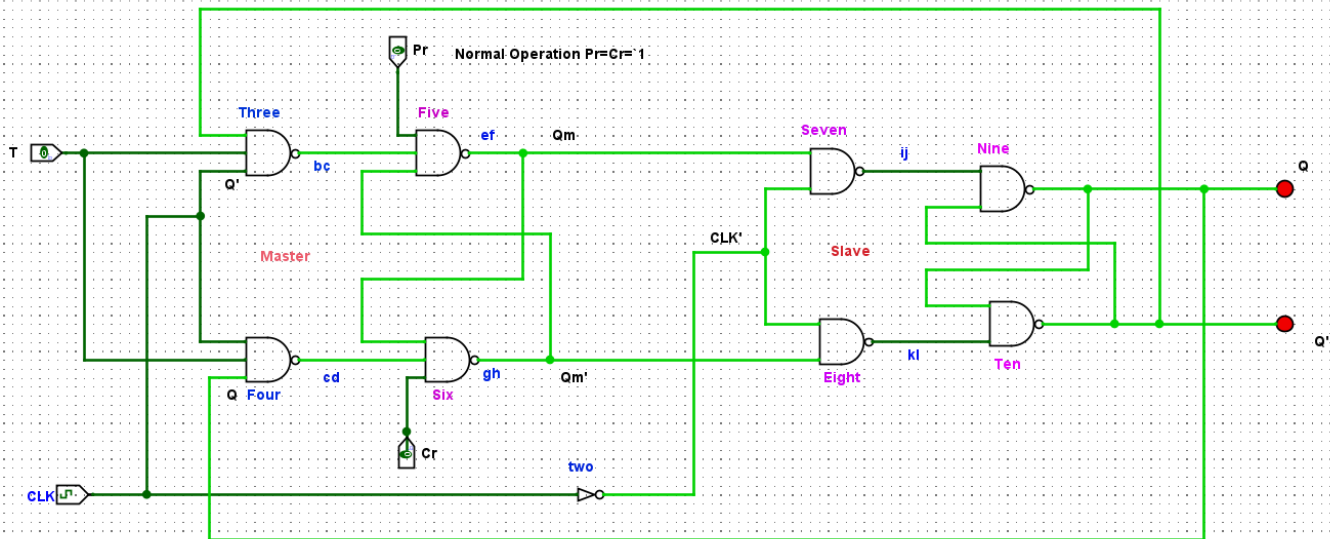




T	Q	Q'
0	0	0
1	0	1
0	1	0
1	1	0

Synchronous Preset and Clear:

```

module t_type (
    input wire clk,
    input wire t,
    input wire pr,
    input wire cr,
    output reg q,
    output reg q_bar
);

always @(posedge clk) begin
    if ((pr == 1'b0) && (cr == 1'b1)) begin
        q <= 1'b1;
        q_bar <= 1'b0;
    end

    else if ((pr == 1'b1) && (cr == 1'b0)) begin
        q <= 1'b0;
        q_bar <= 1'b1;
    end

    else if ((pr == 1'b1) && (cr == 1'b1)) begin
        if (t == 0) begin
            q <= ~q;
            q_bar <= ~q_bar;
        end
    end

end

endmodule

```

pr	cr	t	q	q_bar
1	1	0	x	x
1	1	1	x	x
0	1	0	x	x
0	1	0	1	0
1	1	0	1	0
1	1	0	0	1
1	0	0	0	1
1	1	0	0	1
1	1	0	1	0
1	1	1	1	0

Asynchronous Preset and Clear:

```
module t_type (
    input wire clk,
    input wire t,
    input wire pr,
    input wire cr,
    output reg q,
    output reg q_bar
);

always @(posedge clk or negedge pr or negedge cr) begin
    if (!cr) begin
        q <= 1'b0;
        q_bar <= 1'b1;
    end
    else if (!pr) begin
        q <= 1'b1;
        q_bar <= 1'b0;
    end
    else if (t == 0) begin
        q <= ~q;
        q_bar <= ~q_bar;
    end
end

endmodule
```

pr	cr	t	q	q_bar
1	1	0	x	x
1	1	1	x	x
0	1	0	1	0
1	1	0	1	0
1	1	0	0	1
1	0	0	0	1
1	1	0	0	1
1	1	0	1	0
1	1	1	1	0

Testbench

```
`timescale 1ns/1ps

module t_type_tb;

    reg clk, t, pr, cr;
    wire q, q_bar;

    t_type dut (
        .clk(clk), .t(t), .pr(pr), .cr(cr),
        .q(q), .q_bar(q_bar)
    );

    initial clk = 0;
    always #5 clk = ~clk;

    initial begin
        $display("pr cr t | q q_bar");
        $monitor("%b %b %b | %b %b", pr, cr, t, q, q_bar);

        pr = 1; cr = 1; t = 0; #10;
        pr = 1; cr = 1; t = 0; #10;
        pr = 1; cr = 1; t = 1; #10;
        pr = 0; cr = 1; t = 0; #10;
        pr = 1; cr = 1; t = 0; #10;
        pr = 1; cr = 0; t = 0; #10;
        pr = 1; cr = 1; t = 0; #10;
        pr = 1; cr = 1; t = 1; #10;

        $finish;
    end

endmodule
```